

KEY BATTLE

BATTLE OF VERDUN
1916

In February 1916, the Germans attacked the French trenches at Verdun, firing over a million shells in their preliminary bombardment. A battle of attrition followed, in which more than 500,000 men died, with little gain to either side.



▲ French soldiers advance across no-man's-land, in a 1928 film recreating the infantry assault at Verdun.

▼ PASSCHENDAELE

Australian soldiers advance across duckboards during the Battle of Passchendaele, on the Western Front, in 1917. Artillery shells and wet weather combined to create nightmarish fighting conditions, with movement hampered by thick mud and deep, water-filled craters.

KEY DEVELOPMENT

TRENCH WARFARE

World War I (1914–18) was industrialized conflict on a massive scale, with factories supplying weapons and munitions for armies totalling millions. Trench warfare, born out of this new, large-scale firepower, favoured the defenders, especially if they were protected behind barbed wire.

By 1915, a double line of trenches stretched across France and Belgium, with troops facing one another across no-man's-land. This new kind of field fortification was a response to increasing artillery and infantry firepower, as was the adoption of steel helmets by all armies. Trenches developed into elaborate defence systems: the Germans in particular constructed complexes of concrete bunkers and concealed machine-gun nests that extended for miles behind the front line (see pp.286–87).

ATTACK AND DEFENCE

Offensives in the initial stages of trench warfare involved a sustained preliminary artillery bombardment – designed to destroy enemy defences by sheer quantity of high-explosive shells – followed by a frontal infantry assault. Often, however, the defenders survived the bombardment, and slaughtered the attackers as they tried to cross no-man's-land, using machine-

guns and rapid-fire rifles. Even when enemy front-line trenches were captured, assaults floundered as the attackers pushed further forward over ground that had been churned up by shells. The advancing troops could no longer communicate with their own lines by field telephone, with the result that the defenders were often able to move reserves to a danger spot quicker than the attackers could reinforce their offensive. Cavalry units, the only forces capable of rapid movement, were highly vulnerable to infantry and artillery fire, and hampered by barbed wire and shell craters.

BREAKING THE DEADLOCK

Commanders began to seek technological solutions to address or overcome the stalemate of trench warfare. The Germans introduced poison gas, which was first pumped from cylinders and later delivered by artillery shells. Used by both sides,

